

Colby Abuhoff

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Education

Florida Atlantic University, Boca Raton, FL

August 2023 - Present

B.S. in Biomedical Engineering

Expected May 2026

GPA (Cumulative, Unweighted): 3.813

President's List (Fall 2024), Dean's List (Fall 2023, Spring 2024, Spring 2025)

Doral College, Doral, FL

June 2019 – May 2023

Associate's Degree

Summa Cum Laude | GPA: 4.00

Key Skills

Technical Skills: Python | MATLAB | Arduino IDE | SolidWorks | Microsoft Office Specialist: Expert | LabVIEW

Relevant Coursework: Genetics | Dynamics | Circuits | Differential Equations | Bio Signals/Digital Filter Design | Robotics | Control Systems | Biomaterials | Bioimaging | Strengths of Materials | Biomedical Instrumentation

Projects

Lumbar Exoskeleton

September 2025 – Present

- Led the team in creating a CAD model for our adapted SuitX exoskeleton, running FEA to validate the design.
- Designed the exoskeleton to adapt to user movements during recovery.

Epi-Meter

August 2025 – Present

- Designed an epinephrine injector utilizing standard 0.3 mg dosages intramuscularly and subcutaneously to enhance concentration and efficiency.
- Accepted to Florida Atlantic's WAVE competition to further develop the product.

Safe Sense Stick

January 2025 – May 2025

- Created a walking stick for visually impaired people that senses temperature above 80°F and objects up to 4 meters ahead.
- Coded the walking stick to make a buzzer sound higher for temperatures too hot, and vibrate more the closer objects ahead are.

SafeTurn | Infant Car Seat Design

August 2024 – December 2024

- Created a design of an infant car seat that improves comfort of infants in harsh turns with slight adjustments -15° to +15° from home position, correcting harsh head movements.
- Implemented a scented headrest with the purpose of calming the child while in the seat.

Spotify Keychain

June 2024 – August 2024

- Created a keychain that when scanned links to a Spotify playlist, auto-updating with new entries.
- Designed the .98 x 1 x 2.78 (in) fob in SolidWorks.
- Worked with the laser engraver gcode to burn the QR code into a wood block.

Work Experience

Assistant Researcher, Spine Biomechanics Lab, *Boca Raton, FL*

September 2024 – August 2025

- Conducted a systematic literature review about Vertebral Tethering and its impact on Cobb angles.
- Collaborated with a team to discuss important literature topics to specify in the research.
- Utilized Ansys modelling to experiment with varying degrees of Cobb angle changes over time.

Chiropractic Therapist Intern, Integrated Chiropractics, *Boca Raton, FL*

October 2023 – May 2024

- Conducted a variety of therapies such as Stimulation, Ultrasound, Combination, C4 Laser, and Decompression.
- Fixed technical difficulties in machines or software.
- Worked with high-profile figures.

Vice President/Composites Lead, Owls Racing, *Boca Raton, FL*

August 2023 – Present

- Led the team in development of OR-26, taking over as Chief Engineer alongside Vice-President to make executive team design and financial decisions.
- Performed wetlayups and infusion for various parts, utilized precision sanding devices.
- Utilized LiDAR scanning to create a 3D scan of the carbon-infused seat created by the team.
- Created models and drawings in SolidWorks of the OR-25 nosecone and body panels.
- Mixed urethane foams to create a foam reaction that took shape of the driver's back, fitting the seat.

Photographer, Adolph and Levis JCC, *Boca Raton, FL*

June 2023 - Present

- Photographed and worked with children and adults in the Camp Kavod special needs summer camp.
- Created a movie showcasing and advertising funding for the special needs department.